

EMILIA Protocol

The human-authorization receipt layer for agentic AI · proposed for AAIF TC review ·

non-binding feedback sought

MCP governs how an agent **connects**. goose governs how it **executes**. AGENTS.md governs how it's **guided**. EMILIA answers the question those leave open: when an agent's action is **irreversible** — a payment, a vendor-record change, a deletion — what artifact proves a **named human authorized that exact action before it ran?**

MCP how an agent connects	goose how it executes	AGENTS.md how it's guided	EMILIA proof a human authorized the irreversible action
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Decision logs are testimony. Receipts are evidence.

WHAT IT IS

An **offline-verifiable authorization receipt**: a device-bound human signoff (Ed25519 over RFC 8785 canonical JSON) over the **exact** action — verifiable by anyone with open-source code, with **no backend and no vendor trust**. The MCP middleware intercepts an irreversible tool call and returns 428 — no receipt, no execution until a valid, exact-action receipt is presented. Change one field of the action and the signature fails; replay it and it's refused. EP defines **when evidence is admissible for reliance**: signed is not trusted, verified is not accepted, and accepted requires a pinned policy.

IT'S REAL, AND IT'S SMALL

- **IETF Internet-Draft** — draft-schrock-ep-authorization-receipts (posted on the IETF datatracker).
- **Reference verifiers in JavaScript, Python, and Go** — byte-identical agreement across 16 conformance suites / 158 vectors (a cross-language consistency check, one repository — not independent reimplementations).
- **Formally-verified core** — 26 TLA+ safety properties, 0 errors; Alloy models for the quorum path.
- **MCP middleware** that enforces the loop, plus SDKs (@emilia-protocol/require-receipt , /gate , /verify).
- **60-second check**, no account or backend: npx @emilia-protocol/issue demo

PROPOSAL FACTS (FOR THE TC)

License: Apache-2.0 (OSI-approved)	Governance: independent maintainers, IETF-track spec
Repo: github.com/emiliaprotocol/emilia-protocol	Adoption: npm packages live; MCP dev-wedge; outside party (COSA / J Diesel) reproduced our published vectors at a pinned commit
Infra needs: minimal (static site + npm; no backend to verify)	Standards: IETF I-D + cross-language conformance
Composes with: MCP · goose · AGENTS.md · WIMSE	Preferred stage: Growth (or as the TC recommends)

WHAT WE'RE ASKING

Early, **non-binding** Technical-Committee feedback — intake, the Q&A window, or a 10-minute presentation if a member requests one. We understand from the proposal process that any transfer of trademarks or project assets is **Step 6 (Finalize Legal Paperwork)**, executed only after both the TC vote and the Governing-Board vote — so this stage commits nothing. We're seeking guidance on fit, not asking the TC to adopt anything yet.

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Static assessment / reference implementation — not a vulnerability claim. Apache-2.0, offline-verifiable, no vendor lock-in.